DATA SCIENCE · RANDOM FOREST

Cosmetic Risk Detector

Ingredient-based alert prediction with an auditable Random Forest.

Malika Baguari · Ironhack Final Project · October 2026



Made with GAMMA

Can ingredients reproduce a product alert?

THE TASK

Binary

Predict the supplied has_risk label. No independently verified regulatory registry.

THE METRIC

Recall

Prioritize alerts while reporting precision, F2 and average precision.

THE SCOPE

Screening

An educational classification prototype, not a cosmetic safety certificate.

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DATA FUNNEL · OPEN BEAUTY FACTS

From 64,237 raw products to 15,574 for modeling

64,237

Supplied Open Beauty Facts catalog; multiple country tags.

15,621

Supplied labeled input; historical labels preserved.

15,574

Cleaned rows after quarantining 47 low-alignment records.

Intermediate snapshot: 17,354 rows · Three API records saved

The original 16-substance roster and labeling code remain unavailable.



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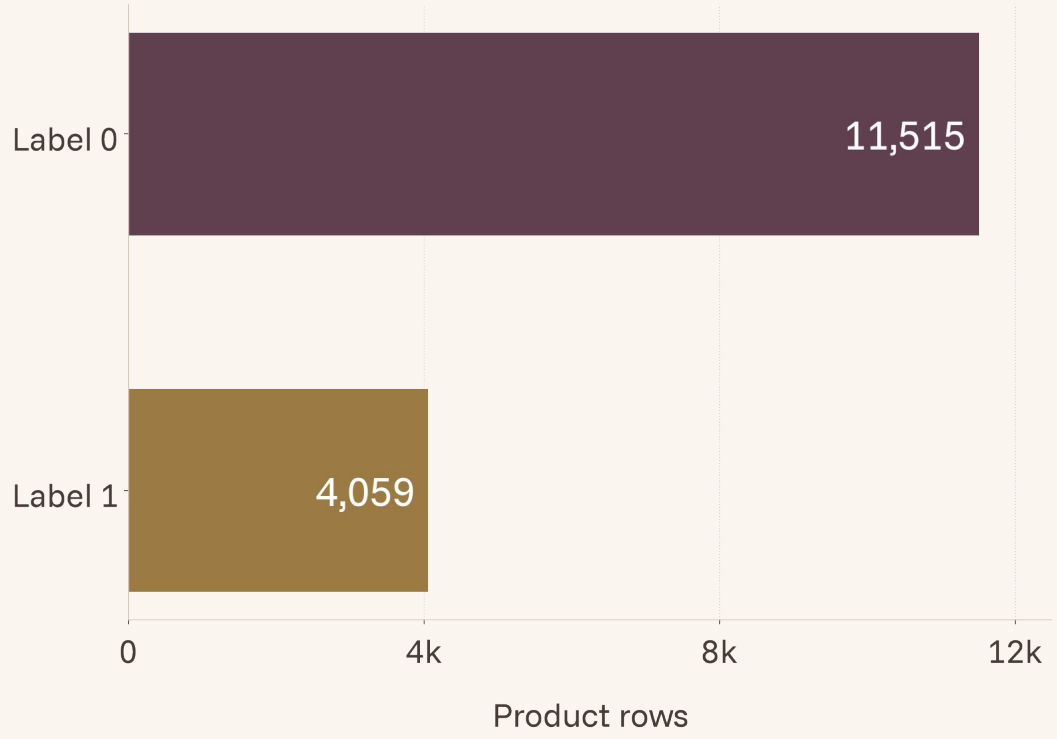
Cleaning with traceable decisions

Normalize Unicode and HTML; safely parse lists. Keep source fields and supplied labels.

Cleaning decision	Rows left	Quarantined
Supplied labeled input	15,621	—
Parse lists; validate labels / codes	15,621	0
Quarantine low text/list alignment	15,574	47
Group repeated formulas; retain rows	15,574	0

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A 74% baseline misses every alert



15,574 cleaned rows; supplied labels, not verified danger.

26.06%

Positive supplied labels: 4,059 product rows.

19.53

Mean ingredient-list length; median 18 entries.

742

Repeated-formula rows; grouped splits.

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EVALUATION DESIGN

Formula-grouped evaluation

TRAINING ROWS

12,459

SHARED TRAIN / TEST FORMULAS

0

Separate contamination from label provenance.

Repeated formulas never cross the split. Ingredient-derived labels do not establish independent cosmetic safety.

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A reproducible four-step pipeline

Training-only vocabulary, grouped CV and out-of-fold threshold selection.

01

Parse

Normalize ingredient lists and preserve supplied labels. No target-substance masking.

02

Encode

Top 1,500 training-fold ingredient indicators plus log1p ingredient count.

03

Forest

Compare models; tune Random Forest with 3-fold grouped CV and four settings.

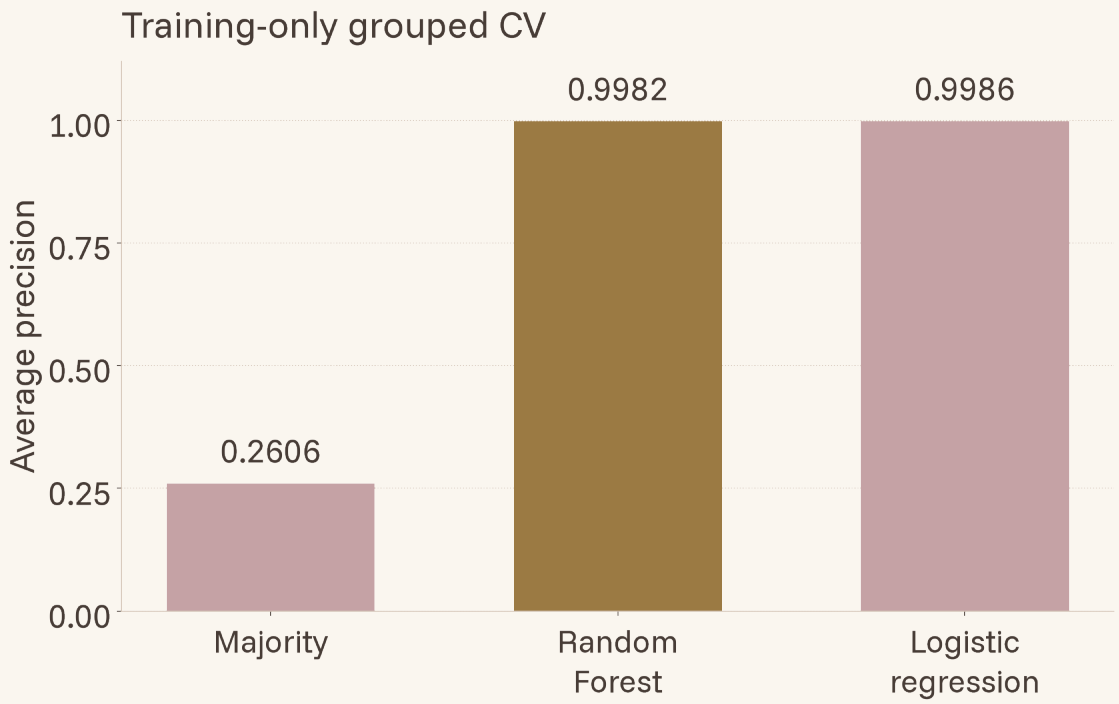
04

Threshold

Choose F2 from training OOF predictions, with a recall target of at least 95%.

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Random Forest: 99.63% recall on the grouped holdout



RANDOM FOREST · HELD-OUT RESULTS

Recall 99.63% · Precision 100.00%

F1 99.81% · Average precision 0.9985

CONFUSION MATRIX · THRESHOLD 0.4387

True negatives: 2,303

False positives: 0

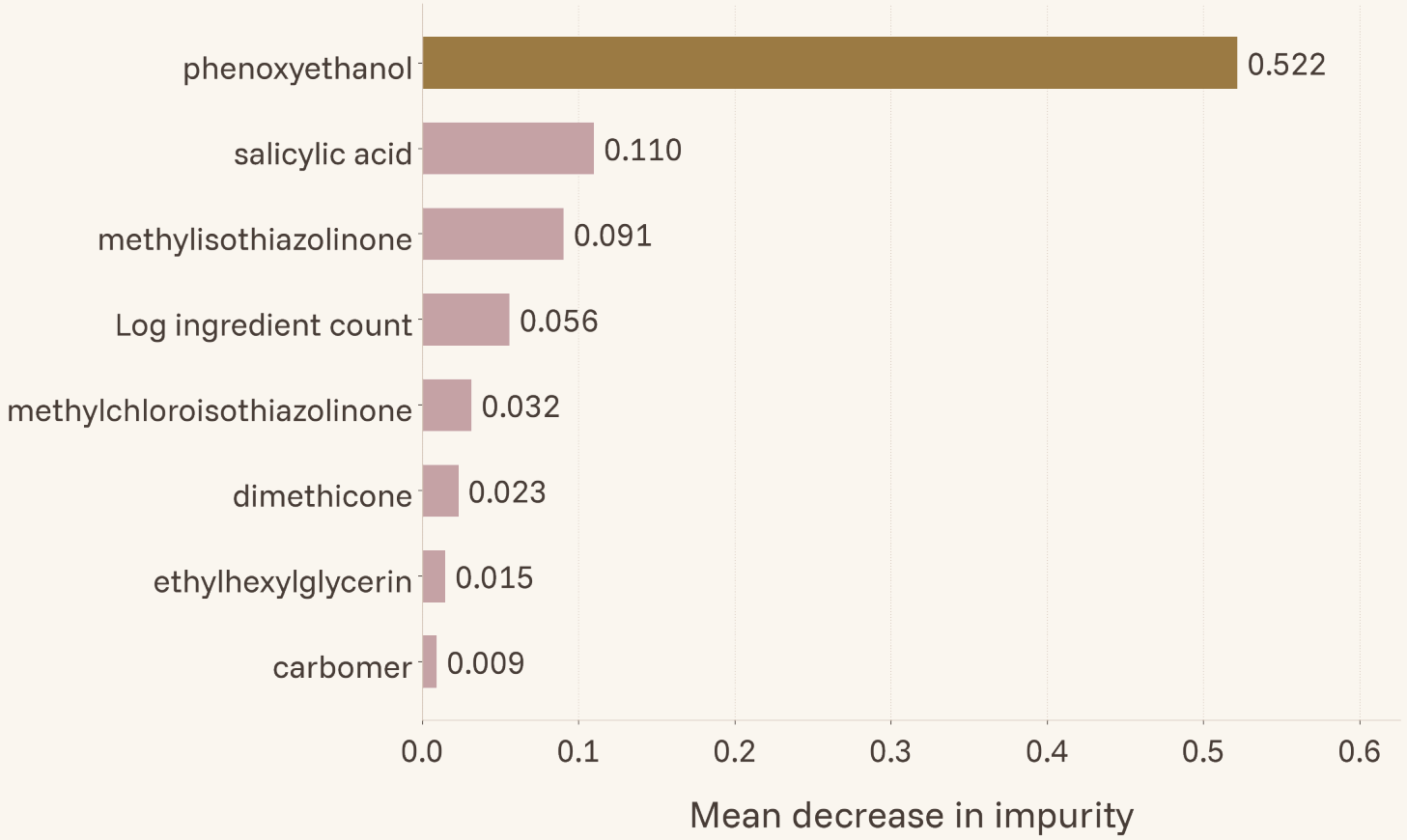
Missed supplied alerts: 3

Detected supplied alerts: 809

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What the Random Forest learns

Actual impurity importance is non-negative and does not indicate toxicity or effect direction.



Top global features

phenoxyethanol · salicylic acid

Not a direction

Importance is not a positive or negative safety effect.

Local explanation

Drop-one-feature changes depend on the full input; they are not causal evidence.

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LOCAL DEMO · RESPONSIBLE USE

A usable prototype with transparent limits

LOCAL STREAMLIT DEMO

Loads the trained forest, predicts supplied alerts and displays local feature perturbations.

REFERENCE RULE: NOT VERIFIED

The 16-substance roster is missing. No deterministic accuracy or masked study is claimed.

Limits: no concentrations, calibrated health probabilities, OCR or independent external validation.

Next: recover the registry, compare deterministic matching and test on independent labels.

Thank you. Questions?



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